

...Infinity in a grain of sand,
And eternity in a flower.

Now you're probably wondering what *this* writer's been smoking, so without much ado, look on this month's CD to find fractal programs such as Fractal Forge (FF) and Fractint. Neither of these require installation. FF, as well as the DOS version of Fractint, feature colour cycling—very, very trippy—while the Windows version of Fractint doesn't support it in most cases. Fractint is the older, most well-known fractal-generating program; FF is newer, and the UI is more user-friendly with more obvious options.

Creativity 101

Fractals are, generally speaking, so intricately beautiful that it's easy to think you're being creative when you generate a sufficiently complex one!

The fact is, it's nature doing it for you—it's numbers working their magic. Now what's hard is coming across something unique—like a fractal island or river in uncharted territory—which comes only with patience, hard work, and all the other virtues.

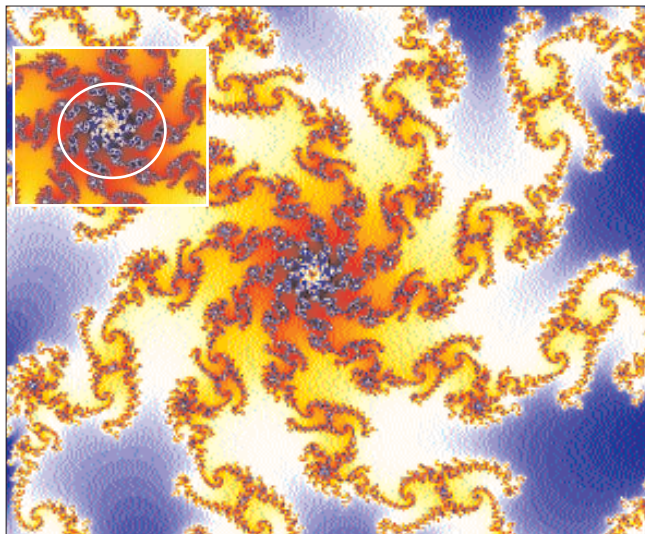


Figure 1: An incredibly elaborate spiral (top left), when zoomed into (the white circle), gives more of the same... there's no getting to the centre!

Right. So when you fire up, say, Fractal Forge, you get an interesting-looking structure. This is called the *Mandelbrot Set*. It's a fractal. Zoom in, and the plot thickens. The pattern gets increasingly complex. Zoom in even more, and you'll notice that "deep inside" is the same shape you saw in the beginning—in fact, lots and lots of them! Tiny little buggers, representative of the whole. Swirling round them are rings, rivers, beaches, islands—all swathed in passionate, psychedelic colour.

Taking Control

We're not going into the mathematics here, but there are a few things you need to know. First read box *How It Happens*, then come back. Good. Now, the basic Mandelbrot set is generated without any special functions; it's just



Figure 2: Hills, islands, rivers, and oceans—the Phoenix set

squaring and adding. Twists happen when you add functions such as the sine and the cosine to the equation.

However, if you follow the rule of continually squaring and adding, you're still in the Mandelbrot "family" of fractals. There are many more—explore the other sets in, for example, Fractint, which are generated using different formulas. Remember that the key attribute of a fractal is that it is self-repeating.

Happening Places

Now that you're hooked, you need to go out and look for where the interesting patterns are. Let's use the standard Mandelbrot set.

Generally, you need to look at two kinds of things, which are usually at the intersection of two major dark areas:

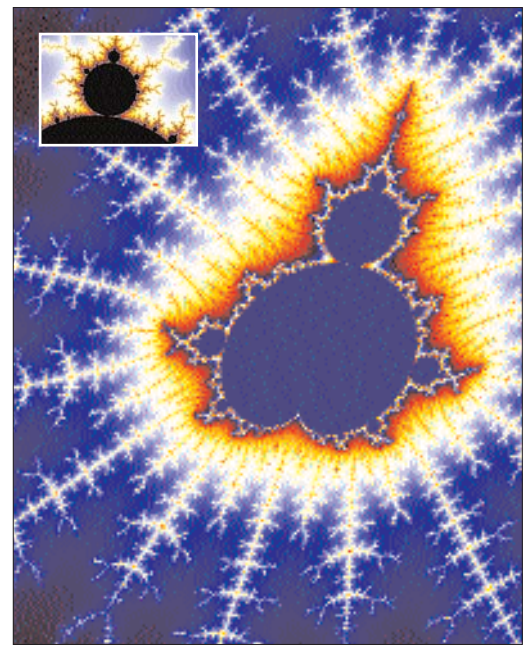


Figure 3: The typical Mandelbrot Shooting Star (upper left), which can be seen in many forms in the Mandelbrot set, shoots into baby Mandelbrot sets like this one. Note that this is typical of an "explosive" landscape